NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

Department of Mathematics

Mid Semester Examinations I

B Tech IV Semester (EEE A & B)

MAIC 203: Numerical and Statistical Methods

Date: 22-02-2024 **Time:** 10.00 AM - 10.50 AM **Max. Marks:** 15

Instructions:

Answer **ALL** the questions.

Use of Calculators is Mandatory.

Start writing your answer from the first page. Do not leave it blank. No extra answer sheet will be provided

Q1. Calculate the approximate value of $4^{\frac{1}{3}}$ using Regula-Falsi method with error tolerance of 0.01. 3M

Q2. Perform three iterations of Gauss Jacobi method to find the solution of the linear system of equations $4x + y + z = 2$, $x + y + 4z = 5$ and $x + 4y - z = 3$. 4M

Q3. The distance $x(t)$ covered by a runner from a fixed point is measured (in meters) at intervals of half a second, starting from $t = 0$. Use Newton's forward difference method to find the distance covered in 0.25 seconds. Also find the velocity of the runner at 1 second. 4M

$x(t)$ in meters	0.00	3.65	6.80	9.90	12.15
	0	0.5	1		

Q4. Compute the value of $\int_0^2 \sin(\pi x)\, dx$ using Gauss Legendre one point, two point and three point formulae. 4M